

Introduction to Computer Vision (83551) - Assignment 3

This assignment is designed to help you achieve several key learning objectives. You will work with PyTorch to build and understand advanced neural network architectures for various computer vision tasks.

Core Objectives of the Assignment

- **Mastering Transformers for Captioning:** You will construct Transformer models from the ground up and integrate them with Convolutional Neural Networks (CNNs) to build a system that can automatically generate descriptive captions for images from the COCO dataset .
- **Applying Self-Supervised Learning:** The assignment will guide you through using self-supervised pre-training methods as a strategy to boost the accuracy and effectiveness of image classification models .
- **Building Generative Diffusion Models:** You will implement and gain a deep understanding of Denoising Diffusion Probabilistic Models (DDPMs), a powerful class of generative models, and apply your implementation to the task of creating novel images .
- **Exploring Advanced Visual Representation Learning:** You will implement two state-of-the-art self-supervised learning frameworks, CLIP and DINO . This will involve learning how these models use large, unlabeled datasets to develop rich and meaningful visual representations .

Question 1: Image Captioning with Transformers

For this task, please follow the steps outlined in the file `Transformer_Captioning.ipynb`. This task includes filling in the code inside:

`transformer_layers.py`

`classifiers/transformer.py`

Note: with the free resources (using GPU T4) plan the code should run in about 3 minutes.

Question 2: Self-Supervised Learning for Image Classification

For this task, please follow the steps outlined in the file `Self_Supervised_Learning.ipynb`. This task includes filling in the code inside:

`simclr/data_utils.py`

`simclr/contrastive_loss.py`

`simclr/model.py`

`simclr/utils.py`

and also, code and inline questions inside `Self_Supervised_Learning.ipynb`

Note: In `Self_Supervised_Learning.pynb` (using GPU T4) the total running time should be around 10 minutes.

Question 3: Denoising Diffusion Probabilistic Models

For this task, please follow the steps outlined in the file `DDPM.ipynb`. This task includes filling in the code inside:

`gaussian_diffusion.py`

`unet.py`

`ddpm_trainer.py`

and also, code and inline questions inside `DDPM.ipynb`

Note: Total running time (using GPU T4) is about 10 minutes.

Question 4: CLIP and Dino

For this task, please follow the steps outlined in the file `CLIP_DINO.ipynb`.

Fill in the code inside the file: CLIP_DINO.ipynb.

Note: Total running time is about 5 minutes.

Important Note:

The following IPython magic commands are intended for local Jupyter environments to auto-reload changed modules:

```
%load_ext autoreload
```

```
%autoreload 2
```

In Google Colab this autoreload extension may not be available or compatible, which can cause errors when running the notebook. **Please delete or comment out** the two lines above so the code will run properly in Colab.

Required Submission Files:

Prepare the following files:

- After you finish all questions open the file `collect_submission.ipynb` inside the `assignment3` directory and run it. It will generate a zip file named: `a3_code_submission.zip`. Upload that file to the VPL on the course website (Lambda).
- In addition, you should create a public GitHub project with all the files inside `assignment3`. Create a link to that project and paste that link in a .txt file. Name that txt file: `[YourName_ID.txt]` and submit that file to Assignment 3.

The submission must include written answers (inside ipynb files), all required uploaded files, and the corresponding code, so that every component of the assignment can be reviewed and fully graded.

VPL Submission System

Submit your assignment through the VPL module in Moodle (Lambda), our dedicated platform for programming tasks.

Academic Integrity Checks

VPL includes a built-in plagiarism detection tool that compares submissions within the course—and optionally across prior semesters—to identify similarities and ensure originality. Instructors review flagged cases manually for fair assessment, upholding academic honesty standards.